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M 207 CIRCULATING WATER SCREENS

1. SCOPE OF WORK

The Bidder shall supply the following for each Block of Muara Tawar Combined Cycle Power Plant Block 2, 3 & 4 Add-On in accordance with **Part 4 Subsection 1.2, “Summary of Work”**

- Two (2) Screen Water washing system each block complete with pumps, valves, piping, spray nozzles and appurtenances.
- Two (2) bar screens each block with one mobile motor driven rake.
- Two (2) power operated double flow traveling band screens each block.
- Trash water trough.
- One (1) trash basket
- Automatic control equipment for the above items.
- All instrumentation and devices for the above items.
- All electrical equipment for all above items.
- All other accessories, bolts and parts embedded in concrete.
- Cathodic Protection for screens, wash pumps and other accessories subject to sea water corrosion.

2. CODES AND STANDARDS

The Bidder shall comply with the requirements of **Part 4, Subsection 2.5 “Codes and Standards”**.

The codes and standards of the following organizations extracted from the **Part 4, Subsection 2.5 “Codes and Standards”** or international equivalent standards are specifically applicable to the design, manufacture and testing of the equipment included in this specification.

AISI	American Iron Steel Institute
ANSI	American National Standards Institute.

ASTM	American Society of Testing and Materials.
AWS	American Welding Society
BS	British Standard
DIN	German Institute for Standardization
EN	European Committee for Standardization
IEC	International Electrotechnical Commission.
JIS	Japan Industrial Standard

3. DESIGN AND OPERATING CONDITIONS

The screens shall be used for screening the main cooling water utilized for cooling the main condenser, closed cooling water heat exchanger, and supplying the desalination plant and chlorination plant.

They shall be designed for outdoor installation and shall be set in a concrete intake structure.

Coarse screens inclined or vertical bar screens shall be supplied and installed across the inlets to the screen chambers complete with an automatic trash rake cleaning system.

Each screen shall be capable of handling 125 % of the total flow of one circulating water pump and screen wash pumps and one desalination feed water pump.

The screen shall perform at guaranteed capacity under the following conditions:

- (a). Sea water (Level, Temperature, Analysis), refer to **Part 4, Subsection 1.1 “Description of Project”**.
- (b). The traveling band screen mesh size shall be in the range between 3 and 6 mm according to Bidder design and calculations and shall be subject to Owner’s approval.
- (c). Maximum allowed pressure drop across screen: 1,000 mm of water.

- (d). The rotation of the screen shall have a high speed and a low speed setting.
- (e). The Trash Rack bar spacing shall be in the range between 20 and 50 mm according to Bidder design and calculations and shall be subject to Owner's approval.

The bar screens rake and traveling band screen shall be operated automatically either by the signal relay of differential water level, timer control or manually.

The pump and screen shall be able to operate independently of each other. Screen water washing pump, self cleaning multi-cartridge type strainer, piping, spray pipe on each screen, water jets, trash water trough and a trash straining basket shall be included.

Two wash water pumps shall be supplied, each capable of delivering the full water requirement for the two screens. The screen water washing pumps shall be vertical spindle, single stage, centrifugal type, one pump shall be on stand by status.

Provisions will be made for isolation of individual screen chambers of concrete stop logs. The stop log guide shall be stainless steel AISI 316 L, equivalent or better.

The level control system shall be designed to detect the differential in level between the inlet and the outlet chambers.

Special Conveying and cleaning elements shall be foreseen for the screens in order to ensure that the jelly fish do not stick to the machine parts.

4. DETAILED REQUIREMENTS

4.1 Traveling Band Screen and Bar Screen

All parts of traveling and bar screen shall be designed to fulfill the needs and conditions which may be imposed during handling, shipping and erection, as well as the actual operation.

All traveling and bar screens when operating shall be free from excessive noise and vibration.

All traveling and bar screens shall be so constructed that dismantling and repairing is easy.

All traveling and bar screens shall be suitable for all weather outdoor service and seawater service.

The structural design of all parts of the traveling and bar screens, where applicable, shall be adequate to withstand a differential head of **shall be adequate to withstand a differential head of a shall be determine by manufacture** without bending or distorting.

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All traveling screens shall be of two speed type, and speed changing shall be manually possible.

The screen trays shall be of sturdy construction.

Screens shall be effectively sealed at both sides and joined between successive baskets so that running clearance at these points shall not exceed the size of mesh opening.

Shear pins to prevent excess loading shall be furnished, and the pins shall be made easily replaceable. A limit switch shall be provided to alarm when shear pin breaks.

Screen housings shall be able to be easily removed for inspection of the inside.

Inspection windows protected by splash-proof plates shall be furnished on back side of screen housing.

Motors and gears shall be of the weather-proof type.

Dirt shall be washed out by screens jet washing water, and shall be collected by screen plates located at the end of pit. Guide plates shall be installed to prevent dirt from falling into the downstream of the screen.

The construction of traveling screen and bar screens shall be easily divided into several parts so as to enable transport, assembly and repair.

Ladder and grating with handrail shall be installed for the maintenance of the motor and gear.

The non-clogging spray nozzles shall be within the enclosure, and shall be arranged to direct fan shaped overlapping splays across the width of the screen trays for washing on the ascending side.

Spray nozzles for traveling screens shall be of removable type.

4.2 Bar Screen with Rake

Bar screen shall be of inclined type and made of SS 316L. Rakes shall be of motor driven type.

4.3 Guide Channel and Foundation Bolts

Embedded materials, such as guide channel and foundation bolts for screen and pump shall be provided by the Bidder. The guide channel shall be of SS 316L.

4.4 Drive

Each screen drive shall be designed with adequate power to start under a 1,500 mm WC differential at high water level.

Motor drive shall be located at an elevation high enough to prevent flooding when the Circulating Water pumps trip.

Speed reduction shall be accomplished by means of a two speed motor in combination with a suitable helical gear speed reducer of the totally enclosed type, direct connection to the motor by means of a flexible coupling. The gear speed reducer is also acceptable.

Shearing devices or other suitable mechanism shall be provided in the driving mechanism to protect the screens from damage in case of jamming.

4.5 Back Washing

The back washing system shall include a back washing pump, duplex strainer spray pipes, header pipes, valves and trash collection trough.

Back washing pumps shall be of the closed shaft vertical centrifugal type with enough capacity and head to properly perform the cleaning operation. Pump bearing shall be cutless with external, filtered flush water sum the pump discharge.

Spray headers shall be furnished with a clean-out connection on the opposite end to the supply water inlet to permit flushing out of the header.

A high energy back-wash shall project all the trash, whatever its nature, into the collecting trough.

The collected trash shall be retained on the unfiltered water side and back washed into an external trough. The trough will have a single exit and an access port that can be inspected while machine is running.

4.6 Materials

All materials of construction shall be identified with ASTM or AISI standard numbers. All features of design and construction, and all materials, shall be acceptable to the Owner.

The principal parts, such as chains, screen wire mesh, baskets with rakes, rollers, pins, bolts and nuts, shall be made of SS 316, and shall be designed for use under uniform loading and to minimize the wear of parts for long time service. Other parts shall be constructed of or protected by corrosion proof materials.

The mesh shall be of stainless steel interwoven with a wire thickness not less than 2mm.

A duplex strainer shall be installed with a mesh size adequate to avoid plugging of the spray nozzles. It shall be constructed of 316L stainless steel.

Spray pipe and nozzles shall be constructed of 316L stainless steel.

Spray water pumps shall be stainless steel construction. All welded stainless steel shall be "L" type.

Materials shall be suitable for the seawater conditions.

4.7 Housing

Each Screen shall be furnished with a reinforced fiber glass splash proof enclosure above the operating floor covering the entire screen assembly. Each screen housing shall be furnished with suitable watertight internally lighted

observation windows and access doors. The design of the housing and access covering shall ensure no spillage of water onto the operating floor.

4.8 Drive

Each screen drive shall be designed with adequate power to start under a 1,500mm WC between upstream and down stream the screen. Each screen drive shall be capable of operating continuously with a differential head of 1,000 mm WC. The equipment shall be designed to withstand not less than 2,400 mm WC differential head without excessive deflection or injurious effects on any part of the equipment. The motor shall be sized so that breakdown torque will not be exceeded at differential heads of 2,400 mm WC.

Motor drive shall be located at elevation high enough to prevent flooding when Circulating Water pump trip.

Speed reduction shall be accomplished by means of a 2-speed motor in combination with a suitable helical gear speed reducer of the totally enclosed type, direct connected to the motor by means of a flexible coupling.

Shearing devices or other suitable mechanisms shall be provided in the driving mechanisms to protect the screens from damage in case of jamming.

4.9 C.W. Trash Rack and Mobile Rake

The CW trash rack shall be constructed of stainless steel 316 and shall consist of:

- A fabricated removable bar screen, bolted on lugs
- A set of removable guides with buffers (top and bottom)

Bar screens shall be of mechanisms type and rake shall be of motor driven type.

The mobile rake shall consist of:

- A mobile rake with wheel and running on the guide rails to position the rake on the different trash racks.
- A tilting debris collection rake hanging from the hoists, with wheels and axles running on the guides.

- Mobile rake shall be equipped with related equipment to dispose of collected solid material into the dumping container.
- A rake carrying assembly, swivelling on the trolley with swivel axles and tilt roller.
- All electrical equipment complete in all respect including motors, electrical operated brakes, limit switches, control panels, slack rope safety devices and lights.

All components of the trash racks shall be constructed of 316L stainless steel including the bar screens. However the trolley shall be made of epoxy coated carbon steel and hoist drum shall be made of cast iron. Any other materials used shall be acceptable to the Owner.

4.10 Trash Baskets

One trash basket for each unit shall be provided. Each basket shall have at least 3 m³ capacities and shall be fitted to facilitate handling with a movable crane.

5. PERFORMANCE TESTING

After equipment is operational, the Bidder shall run a performance test on the equipment to ascertain that it is performing according to the guarantee in every respect. Any deficiencies brought out by this test shall be promptly corrected by the Bidder.

6. QUALITY ASSURANCE / QUALITY CONTROL (QA/QC)

Quality control shall conform to the requirements of **Part 4, Subsection 2.7**.

The quality control documentation shall include, but not be limited to the following:

- Shop Test Reports
- Material Certifications
- NDE Reports
- All other documentation required by applicable codes and standards.

7. INSTRUMENTATION AND CONTROLS

7.1 General

The Bidder shall be responsible for the design, supply and installation of complete instrumentation and control systems and devices for the Circulating Water Screen including but not limited to those listed and described in this Specification.

All instrumentation and control devices furnished and installed by the Bidder shall be wired by the Bidder to junction boxes furnished and installed by the Bidder.

7.2 Local Control Panels

The Bidder shall furnish and install a watertight NEMA type-4 free standing local control panel with degree of protection IPW 66 according to IEC 529.

The local control panel shall contain all control and monitoring devices which do not need to be installed at a local measuring or control location, and all operator switches and alarms.

The local control panel shall also contain all necessary control system hardware/modules.

All instrument and control devices shall be wired to terminal blocks and piped to terminal points as applicable.

Refer to **Part 4, Section 3, I-002 “Local Control Panels”** for the design requirements for the local control panels.

7.3 Instrumentation

The control system, instrumentation and interface devices for each screen shall include but not be limited to the following:

- Instrumentation to measure the difference between the water levels at the inlet and outlet of the screen. This instrumentation shall be used to start the screen and to select high or low screen speed. It shall also provide local high level alarms, a CW Pump trip signal and inputs to the main control room and plant computer.

- Local/remote and auto / manual selector switches and stop / start switches for the screen and backwash pumps. Start and stop of screen and back wash pumps shall be made possible by one touch system on the Operator Station located in the central control room.
- Interlocking as necessary between screen drive motor and the backwash pump motor. The backwash pump shall be started before the rotation of the screen
- Interlocks to prevent the associated C.W. pump from being started unless the screen is running or under automatic control.
- All necessary pressure gauges and flow indicators.
- All pressure reducing valves and fittings required for compressed air.

8. ELECTRICAL SPEC IFICATIONS

8.1 General

The Bidder shall furnish and install all electrical equipment and accessories required for the Circulating Water Screens.

Those equipment shall include but not be limited to the followings:

- (1) Motors including wiring to local junction boxes.
- (2) Cathodic protection system
- (3) All electrical control, motoring and instrumentation devices and other appurtenances required for this Specification M-207 “Circulating Water Screen ”shall be provided by the Bidder.

For general electrical requirements, see **Part 4, Section 3, E-001 “General Electrical Specification”**.

8.2 Motors

For detailed electrical requirements, see **Part 4, Section 3, E-210** for Low Voltage Motor.

8.3 Cathodic Protection System

For detailed electrical requirements, see **Part 4, Section 3, E-260** for CW system cathodic protection.

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